CAS CS 538. Solutions to Problem Set 10

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How to submit: See instructions on Problem Set 1.

1 Commitments

Problem 1. [El Gamal commitments] We now consider a different commitment than the one from Discussion 10. Let $\mathbb{G}$ again be a group of prime order q generated by g. Consider the following protocol between Alice and Bob. Given a message $m \in \mathbb{G}$, Alice runs the following algorithm:

- <u>Commit(m)</u>: pick random $u \xleftarrow{R} \mathbb{G}$, $r \xleftarrow{R} \mathbb{Z}_q$ and let $c = (u, g^r, u^r \cdot m)$.

Alice sends c to Bob and keeps r to herself. Bob makes sure that he got three elements of $\mathbb{G}$.

Some time later, Bob wants to know what m is and Alice is ready to reveal it. Alice sends r and m to Bob. Of course, she may not be honest, so we will call what Bob receives m' and r'. Bob runs the following algorithm:

- <u>Verify(c, m', r')</u>: First, make sure that $m' \in \mathbb{G}$ and $r' \in \mathbb{Z}_q$. Let $c = (u, v, z)$. Check that $v = g^{r'}$ and that $z = u^{r'} m'$. If so, accept m' as the correct message. Else, reject.

Note that this commitment scheme is just the encryption scheme known as multiplicative ElGamal (Exercise 11.5), but without decryption.[1]

(a) Prove that this commitment scheme is *computationally hiding* under the DDH assumption.

The definition of computationally hiding commitments is essentially the same as the definition of semantically security encryption schemes: given $\mathbb{G}, g$, the adversary outputs m_0 and m_1, gets a commitment to m_b for random bit b and outputs $\hat{b}$, with the usual measure of adversarial advantage $\mathsf{Comadv} = |\Pr[\hat{b} = 1 \mid b = 0] - \Pr[\hat{b} = 1 \mid b = 1]|$.

Solution. Your solution goes here

(b) Prove that the commitment is *perfectly binding*: namely, no matter how clever or computationally unbounded Alice is, for every c and every r_1, r_2, there is at most one m that Bob will accept. In other words, for every c, r_1, r_2, m_1, m_2, if Verify(c, m_1, r_1) accepts and Verify(c, m_2, r_2) also accepts, then $m_1 = m_2$.

Solution. Your solution goes here

[1] In fact, a similar commitment scheme can be built out of any encryption scheme that uses randomness r: to commit to m, output pk and $E(pk, m)$ where E uses randomness r and no other randomness. To verify a revealed message m', use r' to re-encrypt m' under pk and check that the result matches the received commitment. It works as long as for every public key — even adversarially generated — correct decryption is unique.

Problem 2. (20 points) The above scheme is *perfectly* binding and *computationally* hiding. In discussion 10 we covered Pedersen commitments, which are *computationally* biding and *perfectly* hiding.

To recap the discussion, a commitment scheme is *perfectly hiding* when for every g, h, m_0 and m_1, the distribution of outputs produced by $\text{Commit}(m_0)$ is identical to the distribution of outputs produced by $\text{Commit}(m_1)$. (Note that this protects Alice during the commit phase because Bob learns nothing about m, regardless of Bob's computational power or any computational assumptions.)

Prove that it's not possible to have a commitment scheme to be both perfectly binding and perfectly hiding.

Solution. Your solution goes here

2 Weak PRF to PRF

Recall that in the attack game for a PRF F defined over $(\mathcal{K}, \mathcal{X}, \mathcal{Y})$, the adversary $\mathcal{A}$ submits a sequence of queries to the challenger, and the challenger picks a random key k and answers each query x_i with $F(k, x_i)$ in Experiment 0, and it picks a random function f from the set of all functions from $\mathcal{X}$ to $\mathcal{Y}$ and answers each query x_i with $f(x_i)$ in Experiment 1. Note that the adversary has full control over the queries.

We can, however, define a weaker notion of a PRF, called a *weak PRF*, where the adversary is only allowed to make function queries on random inputs. More specifically, in the weak PRF game, whenever the adversary $\mathcal{A}$ makes a query, the *challenger* picks a random $x_i \in \mathcal{X}$ and answers the query with $(x_i, F(k, x_i))$ in Experiment 0, and with $(x_i, f(x_i))$ in Experiment 1. As expected, the adversary's advantage in this game, denoted $\text{wPRF}\mathsf{adv}[\mathcal{A}, F]$, is defined as the difference between the probability that $\mathcal{A}$ outputs 1 in Experiment 0 and the probability that $\mathcal{A}$ outputs 1 in Experiment 1; and the PRF F is said to be a weak PRF if $\text{wPRF}\mathsf{adv}[\mathcal{A}, F]$ is negligible for all efficient adversaries $\mathcal{A}$. (See definition 4.3 in Section 4.4.1 of the textbook if you want more explanations.)

Problem 3. (35 points) Problem 11.1 in the textbook. Correction to the problem: let q be the total number of queries that $\mathcal{A}$ makes. Then

$$\text{PRF}^{\text{ro}}\mathsf{adv}[\mathcal{A}, \mathcal{F}] \leq \text{wPRF}\mathsf{adv}[\mathcal{B}, \mathcal{F}']$$

should be instead

$$\text{PRF}^{\text{ro}}\mathsf{adv}[\mathcal{A}, \mathcal{F}] \leq \text{wPRF}\mathsf{adv}[\mathcal{B}, \mathcal{F}'] + \frac{q^2}{2 \cdot |\mathcal{X}|}$$

Interpretation: We know that $\mathcal{A}$ is polytime bounded and hence it can make only a polynomial number of queries. Thus, q^2 is negligible if $|\mathcal{X}|$ is exponentially large. So this weak-PRF-to-PRF conversion works for large $|\mathcal{X}|$ (something like 256-bit is reasonable) but not for small $\mathcal{X}$.

Approach: Let $\mathcal{A}$ be a PRF adversary against F. Because we are in the random oracle model, $\mathcal{A}$ gets to ask two types of queries:

- q_p PRF queries m_i (for $1 \leq i \leq q_p$) to the PRF (in Game 0) or random function (in Game 1)

- q_H random oracle (RO) queries $\hat{m}_i$ (for $1 \leq i \leq q_H$) to H

The total number of queries is $q = q_p + q_H$.

Let p_0 be the probability that $\mathcal{A}$ outputs 1 in experiment 0 of the PRF game (queries answered by F), and p_1 be the probability that $\mathcal{A}$ outputs 1 in experiment 1 of the PRG game(queries answered by a random f).

Let $\mathcal{C}$ be the following challenger: pick a random function $H : \mathcal{M} \to \mathcal{X}$ and a random function $f' : \mathcal{X} \to \mathcal{Y}$; answer PRF queries m_i with $f'(H(m_i))$ and RO queries $\hat{m}_i$ with $H(\hat{m}_i)$. Let $p_{\mathcal{C}}$ be the probability that $\mathcal{A}$ outputs 1 when interacting with $\mathcal{C}$.

In the text two parts, you will prove $|p_0 - p_c|$ and $|p_c - p_1|$ are both small, giving you that $|p_0 - p_1|$ is small.

(a) Prove that $|p_0 - p_c| \leq \text{wPRF}\mathsf{adv}[\mathcal{B}, \mathcal{F}']$ by building an adversary $\mathcal{B}$ that uses $\mathcal{A}$ and plays the wPRF game of F' with advantage $|p_0 - p_c|$.

Hint: $\mathcal{B}$ has to figure out how to consistently answer RO queries of $\mathcal{A}$. For example, if $\mathcal{A}$ asks $\mathcal{B}$ for $H(m)$, $\mathcal{A}$ may later ask $\mathcal{B}$ for $F'(H(m))$ (or $f'(H(m))$, depending on which game it's in). $\mathcal{B}$ should prepare for this eventuality, because it can't ask its wPRF challenger for a particular input. (This is called "preparing" or "programming" the random oracle.)

Solution. Your solution goes here

(b) Prove that $|p_c - p_1| \leq \frac{q^2}{2 \cdot |\mathcal{X}|}$ by the difference lemma.

Solution. Your solution goes here